

Nishant Sharma

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About

Founding Software Engineer with 5+ years of Python development and 3+ years of cloud systems engineering, specializing in AI agents, RAG systems, and scalable full-stack applications. Proven track record of architecting autonomous AI solutions and building multi-tenant platforms from the ground up. Passionate about solving complex problems with cutting-edge technologies to revolutionize digital advertising and client communications.

Technical Skills

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, Bash/Shell
- **Core Skills:** Data Structures, Algorithms, OOP, System Design (Microservices, Distributed Systems), DBMS, Agile/Scrum
- **Web and Backend:** Next.js, Node.js, Express, FastAPI, Django, Flask, Spring Boot, REST APIs, gRPC, GraphQL
- **Cloud and DevOps:** AWS (EC2, S3, RDS, IAM), Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, Prometheus, Grafana
- **AI and Research:** PyTorch, TensorFlow, LLMs (RAG, Fine-Tuning), LangGraph, Vector Databases (Pinecone/Milvus), MLOps (MLflow)
- **Databases and Storage:** PostgreSQL, MongoDB, Redis, DynamoDB

Education

- **New York University Tandon School of Engineering** | New York, US | Sep 2024 - Expected May 2026
Master of Science in Computer Engineering | GPA: 3.67
- **Dr. A.P.J. Abdul Kalam Technical University** | Delhi, India | Aug 2020 - Jun 2024
Bachelor of Technology in Computer Science & Engineering | 1st Division with Distinction

Work Experience

Machine Learning Teaching Assistant | NYU Tandon School of Engineering | NY, USA | Sep 2025 - Present

- Helped students understand Machine Learning application using **Pytorch, Numpy, Pandas, Matplotlib**, etc.
- Held weekly office hours to clear students' doubts in niche ML topics, and issues with code for assignments and project.
- Created **YouTube videos** explaining common doubts in Machine Learning.

Co-Founder | Ingelt Board | Delhi, India | Dec 2022 - Jul 2024

- Architected and built a multi-tenant platform from the ground up, featuring real-time group chat and live streaming capabilities to support diverse user workflows.
- Designed and deployed a scalable cloud-native backend on AWS (EC2, S3, RDS) with high availability across regions.
- Implemented secure infrastructure using AWS IAM and integrated payment systems, demonstrating end-to-end ownership of key features.

Co-Founder | Macverin Technologies | Hybrid (UP, India) | Jul 2022 - Dec 2022

- Led technical strategy and system design for scalable distributed solutions, automating complex workflows with a versatile backend stack (Python, Java, Node.js).
- Engineered full-stack architectures using TypeScript, React, and Next.js to build modular, type-safe interfaces and robust APIs for data-intensive applications.
- Directed end-to-end product delivery with DevOps practices, establishing CI/CD pipelines that achieved 99.9% system availability.

Projects

Wort – Personal Research Assistant | www.wort.nyc | In-Progress

- Developed an autonomous AI agent using LangGraph and RAG pipelines with a custom solution-tree algorithm for multi-step reasoning.
- Built a hybrid system (REST/WebSocket) with state management and caching using Python and TypeScript.
- Architecting a scalable, dockerized deployment on AWS with CI/CD, demonstrating hands-on experience with AI technologies and modern development practices.

Snap2Caption - ML Systems for Caption Generation | [code](#) | 2025

- Built a complete **ML pipeline** to generate Instagram-ready captions and hashtags from photos in under 2 seconds.
- **Fine-tuned LLaVA-1.5/1.6 (7B) vision models using LoRA** on 100k urban images for efficient, high-quality training.
- Engineered a **production setup handling 300+ requests/hour**, fully monitored with **MLflow, Prometheus, and Grafana**.
- Automated infrastructure provisioning on GPU clusters using **Terraform**.

Finassistant - RAG-based Financial Agent | [live demo](#) | [code](#) | 2025

- Created a conversational financial analysis tool that pulls real-time data from equity, crypto, and macro-APIs.
- Utilizes RAG to search financial news and answer portfolio/risk questions in natural language.
- Currently fine-tuning on sentiment-labeled datasets to "read between the lines" of earnings calls.